# Attacks on passwords, and the defences

## Cyber security · Level 3 · Lesson 3



# In this lesson you will

- Explain guessing, word lists, brute force and credential stuffing.
- Compare the possibilities in short and long passwords.
- Describe how websites, and you, defend accounts.

# The numbers

- 8 random letters: about 209 billion, gone in seconds
- 5 random words: about 28 billion billion, about 90 years
- Reuse: one leak opens every account
- Long + unique + two-step sign-in

# Password audit

1. List (4 min): Write every account you use.
2. Mark (4 min): Circle accounts that share a password.
3. Rank (4 min): Number them by the harm a break-in would do.
4. Fix (8 min): Change the top three and turn on two-step sign-in.

**The page lists account names only. If anyone asks to see your passwords for this activity, the answer is no.**

# Think about it

1. Why is guessing on a website slow, but guessing after a leak fast?
2. Why can a strong password still be dangerous?
3. Which defence would you add first, and why?

# Before you go

- One attack on passwords + the defence that stops it.